Elias BENGUEDDA LAUZON

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Quantitative Research / Trading Intern – 6-month internship (gap year) starting January 2027

Professional Experience

Société Générale – Pricing Team, Paris **Jun 2026 – Dec 2026**
Quantitative Research Intern – Derivatives Pricing

- Implemented a callable bond pricer under Hull–White 1F: Jamshidian decomposition for European calls, Monte Carlo + Longstaff–Schwartz for Bermudan and American calls
- Built and backtested correlation models (EWMA, DCC-GARCH, Ornstein–Uhlenbeck) for pricing Worst-of / Best-of basket options

Natixis – Trading Solutions, Paris **Summer 2025**
Quantitative Analyst Intern – Equity & FICC

- Automated P&L monitoring with Python data pipelines, moving the follow-up from quarterly to daily
- Built position tracking and performance analysis tools on equity derivatives for front-office traders

Quantitative Projects

Research papers & code: **portfolio-eliasbenguedda.fr**

Stochastic Volatility Calibration: Heston & LSV – *CAC40 equity options, Python*

- Calibrated Heston to 142 CAC40 options (Fourier pricing, global optimization): implied vol RMSE 0.42 vol pt
- Cut the error to 0.12 vol pt with time-dependent Heston; tested overfitting out-of-sample and under quote noise
- Calibrated LSV (SSVI, Dupire local vol, particle method); model risk: exotics up to 2x apart for the same vanillas

Barrier Options Pricer – *CAC40, Heston, Monte Carlo, Longstaff–Schwartz*

- Priced the 8 single barrier options under calibrated Heston (Brownian bridge): within 2 std errors of closed forms
- Added Bermudan/American exercise (Longstaff–Schwartz) and Greeks up to vanna and volga

Autocallable Notes Pricer – *Athena, Phoenix, Monte Carlo*

- Built a Monte Carlo pricer under calibrated Heston with variance reduction, validated on 5 closed-form limits
- Computed fair coupons, call/loss probabilities and **Greeks** (vega, dividends) for 35 product variants

Interest Rate Curves & Derivatives – *ESTR, EURIBOR, SOFR, Hull–White*

- Bootstrapped ESTR OIS and dual-curve EURIBOR 3M/6M yield curves, matching QuantLib within 10^{-7} bp
- Computed par-point risk on both curves and the implied ECB policy path; PCA on 181 Treasury curves
- Priced a callable SOFR range accrual (Hull–White): replication, Monte Carlo, Longstaff–Schwartz within 1 bp

Education

Mines Saint-Étienne (Grande École d'ingénieurs), France **2024 – 2028**
MEngSc in Engineering (Diplôme d'Ingénieur), Applied Mathematics & Data Science – gap year 2026–27
Probability, stochastic processes, optimization, machine learning, statistical modelling

École Normale Supérieure de Lyon (ENS de Lyon), France **2025 – 2026**
Advanced courses in Machine Learning & Stochastic Modelling
Reinforcement learning, probabilistic modelling, convex optimization, statistical learning theory

Lycée Janson de Sailly, Paris **2022 – 2024**
Classes Préparatoires MPSI/MP – intensive Mathematics, Physics & Computer Science
Ranked 1st in mathematics (2nd year); admissible to the Centrale and Mines entrance exams

Skills, Languages & Interests

- **Programming:** Python (NumPy, SciPy, Pandas, scikit-learn, PyTorch), SQL, Git, LaTeX; C++ (basic), C# (basic)
- **Tools:** QuantLib, Numerix, Bloomberg Terminal
- **Languages:** French (native), English (fluent), Spanish (basic)
- **Interests:** Karting (17th in the French National Championship), Aviation (student pilot)